

1

CIE - I

Module - I

2a)

- Continuous delivery comes after continuous integration where the code after completion is staged into the staging area.

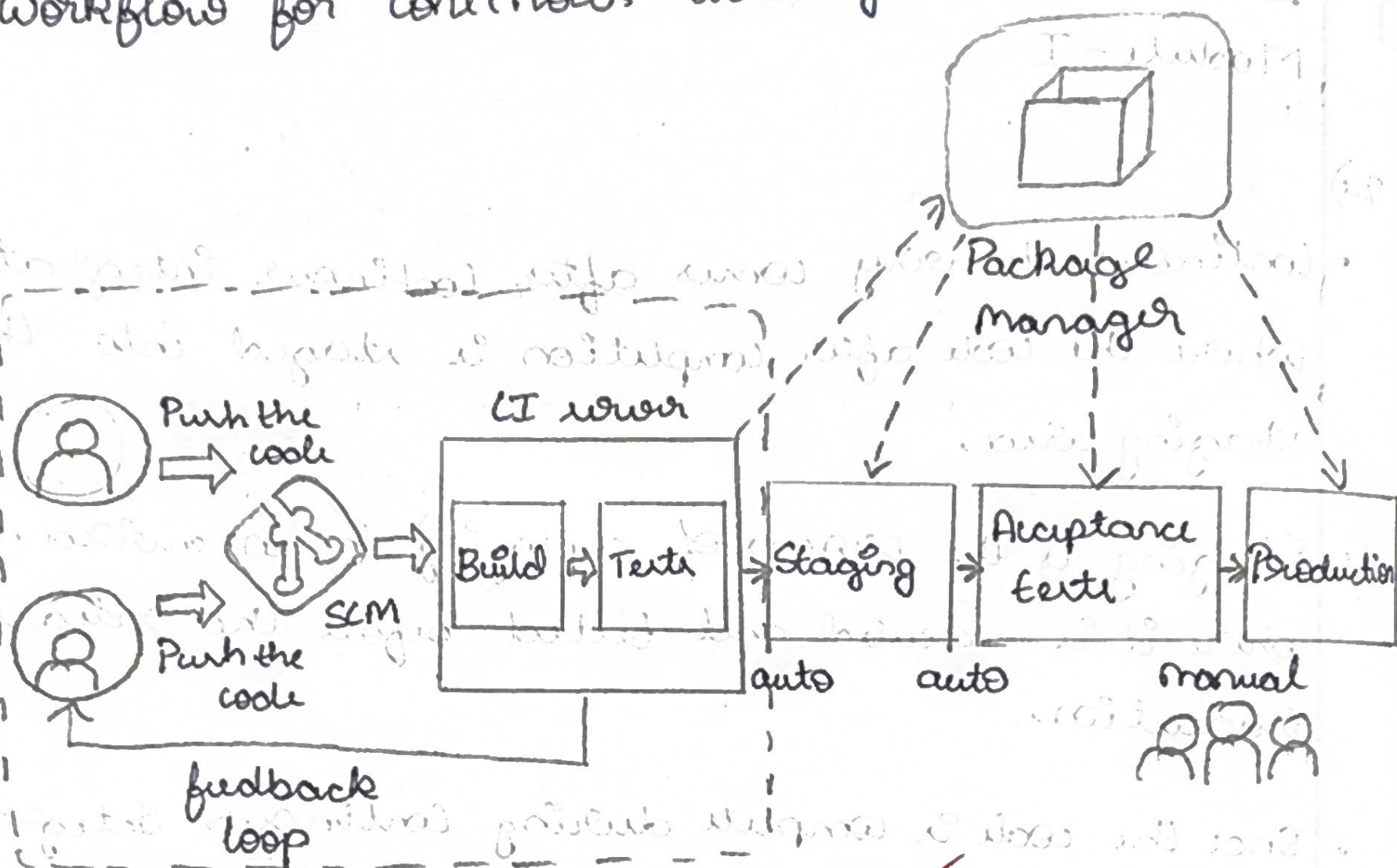
Staging is the process of deploying in an environment where it is executed and tested before the release to production.

- Once the code is complete during continuous integration, it is then staged, then various different acceptance tests are performed if they pass, then it is released to production.

- All these processes are automated except for the step where the product is released to production.

The difference between continuous delivery and continuous deployment is that in continuous deployment all the process including release to production is automated. But in continuous delivery, only the release to production is done manually, the rest are automated.

2. Workflow for Continuous Delivery:



Continuous Integration → Continuous Delivery

- The above diagram shows how Continuous Integration and continuous delivery are connected.
- It also shows how and which process is automated or done manually.
- The whole process of continuous delivery is done with the help of a package manager.

3

2b) IaC stands for Infrastructure as Code. It is a set of practices for writing the code of resources which make up the infrastructure.

- IaC has practices for writing provisioning and configurational code for infrastructure which automates the build and can be used in a repeatable fashion.

- Some benefits of using IaC are:

- i) Standardisation of code
- ii) Code is versioned and controlled
- iii) It can be integrated into the CI/CD pipeline
- iv) Quality, efficiency & speed of code is increased
- v) Project control and management is easier.

- IaC languages and tools have a different way of configuring, there are 3 types:

Scripting

Declarative

Programmatic

i) Scripting Type

It is the type where imperative code must be written in scripts such as azure CLI, bash, powershell etc.

4. Ex: For creating a folder/resource group is called MyAppResourceGroup in the location westusiope, the following two methods can be used:

- Using Azure CLI:

```
az group create --location westusiope --resource-group MyAppResourceGroup
```

- Using Azure PowerShell:

```
New az resource group - name MyAppResourceGroup - location westusiope
```

- So, the following syntax can be used in scripting-type using azure CLI and azure powershell.

2) Declarative type

- It is the type where we can directly mention the state for configuration of infrastructure.

- The languages used here are terraform, typescript etc. Using terraform, the following can be written for the same example of MyAppResourceGroup

resource "azurearm-resource-group" "myrg" {

name = "MyAppResourceGroup";

location = "WestEurope";

tags = {

environment = "bookdemo"

}

• Nginx is a powerful web server which is multi threaded and has event-driven architecture.

• It can also be used for reverse proxy, http caching etc.

• Starting Nginx syntax:

-host: all

worker:

-name: check and install nginx latest version

apt: name = nginx state = started latest

-name: start nginx

service:

name: nginx

state: started

15/15

15/15